From Brains to Models: What AI Reveals About Being Human

Slide-by-slide syllabus companion

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How to use this companion

This teaching guide follows the keynote's 25 slides, one page per slide, with a thumbnail, two reading paths, predictions and exercises. Use it beside the presentation or when revisiting a notebook. The separate perspective paper develops the thematic argument and retains the technical appendices; the tentative solutions work through every slide's exercises for both paths. Each document can be read independently; follow their links when a derivation or a broader discussion helps.

Choose a starting path. The *Biomedical and imaging path* asks what phenomenon needs explaining. It develops biological mechanisms, anatomical relationships, clinical observations and intuitive predictions. Its exercise asks for an observable consequence explained in ordinary language. The *Computational path* asks which objects, assumptions and operations define the problem. It develops variables, domains, units, equations and update rules; its exercise asks for a derivation, calculation, dimensional check, invariant or counterexample. A short *shared interpretation* connects the two results. Choose either first; neither is a prerequisite for the other. The intended audience is MSc and PhD students in biomedicine, imaging and computational science.

Predict, run, explain. State a prediction before opening a notebook. Setup and numerical code are shared: run code cells in order whichever explanation you read first. Change one assumption at a time, compare matched conditions, then explain what the result establishes and what remains untested. Slides 18–20 use Notebook 14d for the transition from anatomical growth to the coupled tumour–oxygen–necrosis model. Its small CPU preset supports exploration; the higher-resolution slide presets are separate choices.

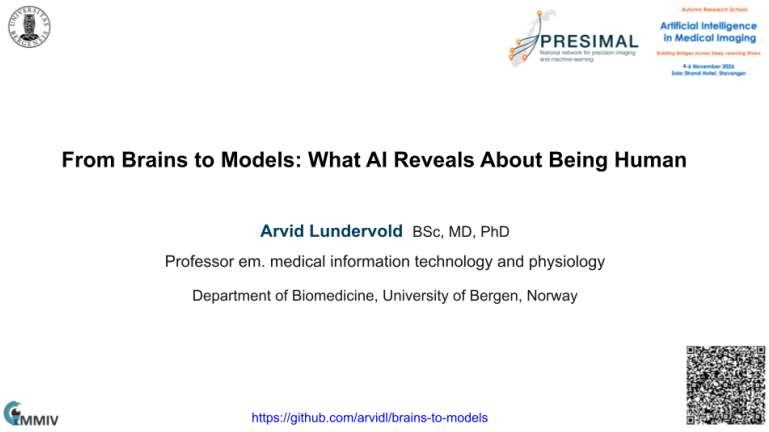
Resources and evidence. The presentation guide maps slides to sources and runnable examples; the notebook guide supplies setup instructions, Colab links and verification scope. Abbreviations, concepts and symbols for both paths are collected in the glossary before the references. Thumbnails follow the presentation; slide 24 uses the credited replacement video-player image and retains the full narration. Model outputs, published observations and interpretive claims are identified separately throughout.

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Slide 01. From brains to models

Biomedical and imaging path

What does it mean when a machine outlines the same lesion as a radiologist? The clinician relates contrast to anatomy, disease history and the person's circumstances. A model could reproduce the contour using a narrower cue. Distinguish the observed agreement, the proposed mechanism and the consequence of using the contour. More reproducible measurement can be useful before a treatment benefit has been demonstrated.

Exercise and result. Choose a lesion-follow-up task. Explain in ordinary language what the contour means, what might mislead it and what a patient would need from its use.

Computational path

Define the objects. Let X be the set of admissible inputs and Y the set of possible outputs. The notation $f : X \to Y$ means that a system assigns one output $f(x)$ to each input x. Let $g : X \to Y$ be another such system. Compare them on a test set $D = \{x_1, \ldots, x_n\}$ of $n > 0$ cases:

$$A_D(f, g) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[f(x_i) = g(x_i)].$$

Read the equation. The index i labels a case. The indicator $\mathbf{1}[\cdot]$ is one if the statement in brackets is true and zero otherwise. The summation counts agreements; division by n gives their fraction. Six agreements among eight cases give $A_D = 6/8 = 0.75$. This number is dimensionless and lies between zero and one.

Interpretation and exercise. Even $A_D = 1$ means agreement only on the tested cases. For $X = \mathbb{R}$, $Y = \mathbb{R}$ and $D = \{0, 1\}$, construct two functions agreeing on D but differing at $x = 2$. State what the new case rules out and why it still does not identify internal mechanisms. Notebook 00 connects this distinction to Marr's levels.

Shared interpretation

The clinical account identifies why the task matters; the counterexample establishes why matching answers cannot identify the process producing them.

From Brains to Models: What AI Reveals About Being Human

Motivation and Inspiration

Slide 02. A learning programme

Biomedical and imaging path

What would you need to understand before relying on an output in your field? Start with an observable phenomenon: a growing lesion, a false-positive screen or a convincing but incorrect report. Describe what is seen, what is inferred and what action might follow. An image boundary, an explanation of disease and a decision have different purposes.

Exercise and result. Predict a result verbally before opening a notebook. After running it, explain a surprise to a colleague using biological or clinical concepts and name an observation that could resolve it.

Computational path

Define the objects. Inputs x belong to a space $\mathcal{X}$ and targets y to $\mathcal{Y}$. An architecture supplies a family $f_\theta : \mathcal{X} \to \mathcal{Y}$, indexed by parameters θ. A loss $L(\hat{y}, y)$ assigns a numerical penalty to prediction $\hat{y}$ against target y. For n observed pairs (x_i, y_i), distinguish

$$\widehat{R}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} L(f_\theta(x_i), y_i), \qquad R_P(\theta) = \mathbb{E}_{(X,Y)\sim P}[L(f_\theta(X), Y)].$$

Read the equations. The first quantity is average loss on the observed sample; the hat marks an estimate. The second is expected loss for cases drawn from population distribution P. Here uppercase X, Y denote random quantities. Expectation $\mathbb{E}$ means an average over their possible values, weighted by probability. Fitting chooses $\hat{\theta} \in \arg\min_\theta \widehat{R}_n(\theta)$: a parameter value achieving the smallest sample loss.

Interpretation and exercise. For zero–one loss, a wrong prediction costs one and a correct prediction zero; risk is an error probability. Give two possible populations compatible with the same finite sample but giving a fixed predictor different risks. Explain why minimizing the observed average does not establish performance in every population.

Shared interpretation

The two routes start with different questions and converge on a claim with an explicit purpose, observable result and boundary of inference.



Slide 03. Biological brains across scales

Biomedical and imaging path

How do cellular organisation and whole-brain form relate to an imaging observation? The slide spans developmental drawings and mammalian brains. Respect the scale bars: drawing size does not determine biological size, and visible branching alone does not measure intelligence. An MRI voxel combines signals from many components; different microscopic changes can have similar macroscopic appearances.

Exercise and result. Describe one contrast directly supported by the image. Then propose a cellular explanation and the additional measurement needed to distinguish it from another explanation.

Computational path

Define the objects. Let $s \in \mathbb{R}^p$ describe a microscopic state and $x \in \mathbb{R}^m$ its measured image features. The measurement model is

$$x = H(s) + \epsilon.$$

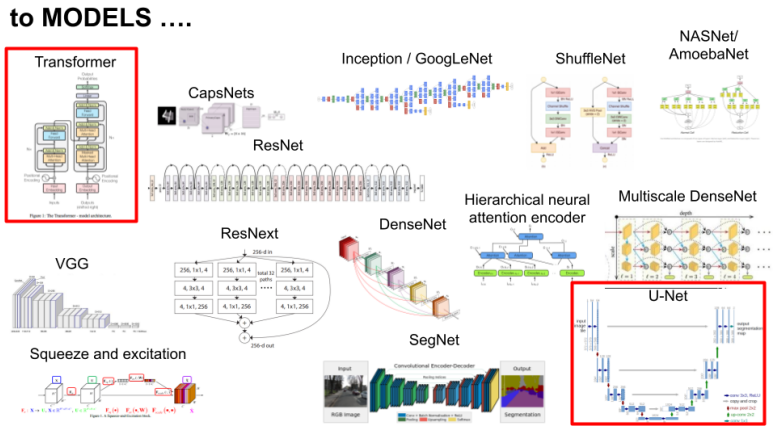
The map H translates state into observations; ϵ represents measurement error in the same units as x. The notation $\mathbb{R}^p$ means a vector with p real-valued components. If H is linear, it is an $m \times p$ matrix, and a vector v in its null space satisfies $Hv = 0$.

Read the consequence. Linearity gives $H(s + v) = Hs + Hv = Hs$. A nonzero invisible change v can therefore alter the underlying state without altering the noiseless observation. For example, $H(s_1, s_2) = s_1 + s_2$ maps both $(1, 2)$ and $(2, 1)$ to 3. Non-uniqueness occurs even before noise is added.

Interpretation and exercise. Add a measurement of the first component and recover both components in this example. State which assumption makes that recovery possible. In Notebook 03, separately identify architecture, encoding and update rule: changing them together leaves their individual effects unresolved.

Shared interpretation

Biological scale gives the observation meaning; the measurement map explains why one observation may support several underlying states.



Slide 04. Architectures encode choices

Biomedical and imaging path

Why does a segmentation task need a different organisation from a whole-image classification task? A contour must preserve location and boundaries; a diagnosis-related score may summarize the image. U-Net combines spatial scales, while Transformers combine representations through attention (Ronneberger et al., 2015; Vaswani et al., 2017). Ask which anatomical relationships and image contrasts the output must retain.

Exercise and result. Explain how a system could perform well on large lesions yet miss small ones. Identify the clinical observation that an average score would conceal.

Computational path

Define the arrays. For image height H, width W and C measured channels, a segmenter maps $f_\theta : \mathbb{R}^{H\times W\times C} \to [0,1]^{H\times W}$. Its output p_i is a score at pixel i; a binary mask uses $m_i = \mathbf{1}[p_i \geq \tau]$ for threshold τ. A score in $[0,1]$ requires calibration evidence before being interpreted as an event probability.

For n tokens, each query and key has d_k components and each value has d_v. Thus queries Q and keys K have shape $n \times d_k$, while values V have shape $n \times d_v$. One attention operation is

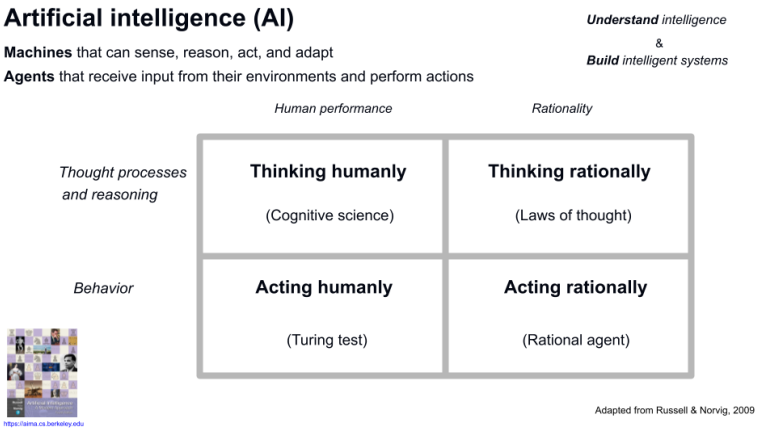
$$A = \mathrm{softmax}(QK^\top/\sqrt{d_k}), \qquad Z = AV.$$

Read the operations. Transposition $K^\top$ exchanges rows and columns; $QK^\top$ is an $n \times n$ matrix. Dividing by $\sqrt{d_k}$ scales its dot-product scores. For scaled score s_{ij}, softmax sets $A_{ij} = e^{s_{ij}}/\sum_{r=1}^{n} e^{s_{ir}}$: exponentiate and divide by the row total. Each row therefore sums to one. Multiplication by V makes each row of Z a weighted combination of value vectors, giving shape $n \times d_v$.

Interpretation and exercise. Keep A fixed and replace V by $2V$. Calculate the new Z and explain why weights alone do not determine the output. Trace the operations in Notebook 01 or 09.

Shared interpretation

Anatomical requirements motivate a design; formal dimensions and counterexamples establish what the design actually constrains.



Slide 05. Four questions about intelligence

Biomedical and imaging path

What kind of success is claimed? The four views separate human-like thought and action from formally rational thought and action. Agreement with historical prescribing could reproduce a systematic mistake. A logically valid argument can start from an incorrect clinical premise. A beneficial action also depends on the person's priorities.

Exercise and result. Place four questions about one diagnostic decision into the grid. Explain what observations each needs and why the same answer could satisfy one question while failing another.

Computational path

Define distinct criteria. For $n > 0$ cases indexed by $i = 1, \ldots, n$, let $\hat{y}_i$ be a model's answer and h_i a human answer. Agreement is $A = n^{-1} \sum_i \mathbf{1}[\hat{y}_i = h_i]$. In formal reasoning, $\Gamma \vdash \phi$ means that conclusion ϕ can be derived from premises Γ using the chosen rules; this does not assert that the premises describe the patient correctly.

For action $a \in \mathcal{A}$, outcome $y \in \mathcal{Y}$ and stipulated utility $U(a, y)$, a rational-action criterion is

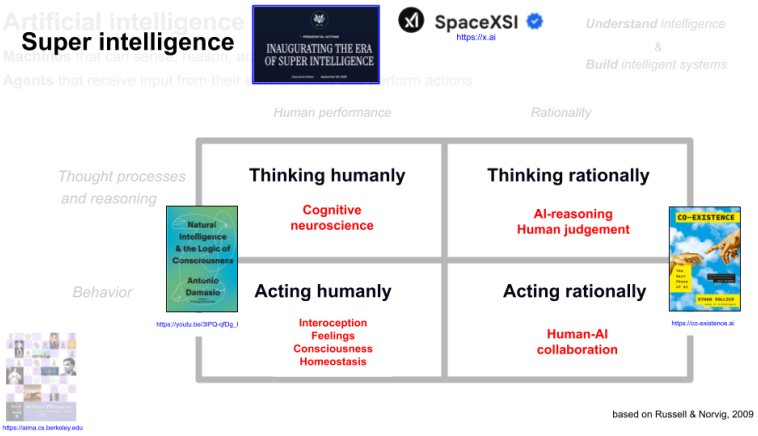
$$a^*(x) \in \arg\max_{a \in \mathcal{A}} \sum_{y \in \mathcal{Y}} U(a, y) P(Y = y \mid x).$$

Read the equation. Conditional probability $P(Y = y \mid x)$ weights each possible outcome after observing x. Summing utility times probability gives expected utility. The operator arg max returns the maximizing action, not the utility value; $\in$ allows more than one action to tie. The displayed sum assumes a discrete outcome set.

Interpretation and exercise. Give two policies whose agreement and expected utility rank them differently. Supply the probabilities and utilities and calculate both rankings. Notebook 04b organizes these evaluation questions; agreement, derivability and utility are different mathematical targets.

Shared interpretation

The clinical route distinguishes meanings of success; the calculation shows that the corresponding objectives can disagree.



Slide 06. Embodiment, judgment and collaboration

Biomedical and imaging path

Why does bodily regulation matter to accounts of cognition? Interoception concerns signals about internal condition; homeostasis concerns maintaining physiological variables within viable ranges. Hunger, pain and fatigue connect perception to biological needs. Their role motivates hypotheses about embodied understanding without providing a sufficient test for consciousness (Seth, 2013; Seth and Bayne, 2022).

Exercise and result. Explain how fatigue could change a person's choice even when the external image is identical. Identify a bodily observation and an alternative explanation for the behaviour.

Computational path

Define a feedback system. At discrete step t, let s_t be the system state, o_t its observation, a_t its action and w_t an external disturbance. A controller uses policy π, possibly with internal memory m_t:

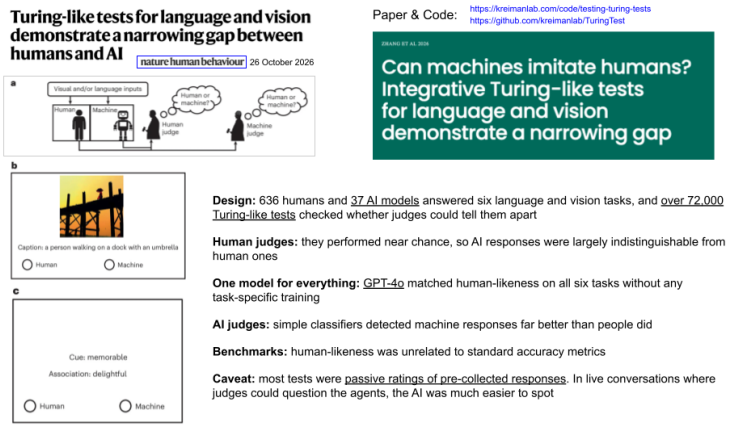
$$o_t = H(s_t), \qquad a_t = \pi(o_t, m_t), \qquad s_{t+1} = F(s_t, a_t, w_t).$$

Read the loop. The observation map H supplies information, the policy selects an action, and transition rule F determines the next state. The subscript $t+1$ means the next update, not a derivative. The set $\mathcal{V}$ contains states declared viable for this model. Invariance means: starting in $\mathcal{V}$, every allowed disturbance and the specified policy keep the next state in $\mathcal{V}$.

Example and exercise. For dimensionless $s_{t+1} = s_t + a_t + w_t$, choose $\mathcal{V} = [-1, 1]$. With full observation, unlimited actions and $a_t = -s_t$, the next state is w_t; hence $|w_t| \leq 0.2$ guarantees invariance. Now limit $|a_t| \leq 0.5$ and allow $|w_t| \leq 0.6$. Exhibit a starting state and disturbance that break the guarantee. This establishes a functional control property, not a criterion for experience; see Appendix C of the paper.

Shared interpretation

Biological regulation explains a meaningful need; the formal controller identifies a testable functional property without equating it with experience.



Slide 07. What a Turing-like test measures

Biomedical and imaging path

What does a person judge in a Turing-like comparison? The cited study's 636 contributors, 37 models and 72,191 judgments concern a specified comparison of human and model responses (Zhang et al., 2026). Fluency, topic and conversational expectations can affect resemblance. Being difficult to distinguish in that setting does not establish the biological process behind an answer.

Exercise and result. Explain how a fluent scan explanation might appear human while omitting uncertainty important to a patient. Propose a comprehension observation distinct from a resemblance judgment.

Computational path

Define one judgment. Let $T_j \in \{0, 1\}$ record item j's true source, with 1 meaning human and 0 model. Judge i supplies classification J_{ij} using the same labels. Correctness is the indicator $Z_{ij} = \mathbf{1}[J_{ij} = T_j]$. If $\mathcal{I}$ is the set of N recorded judge–item pairs, estimated accuracy is

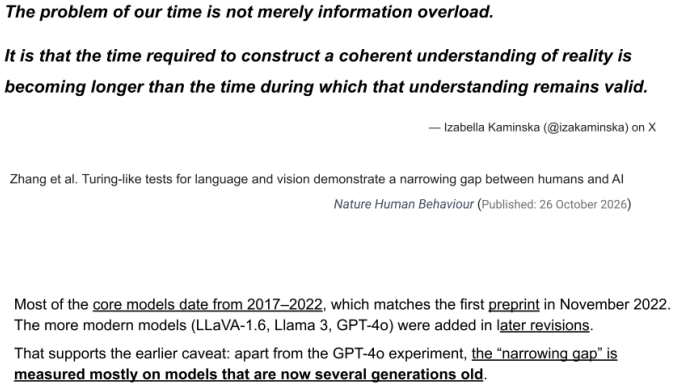
$$\hat{a} = \frac{1}{N} \sum_{(i,j)\in\mathcal{I}} Z_{ij}.$$

Read the uncertainty. For N independent, identically distributed Bernoulli judgments, each with success probability a, $\mathrm{Var}(\hat{a}) = a(1-a)/N$. Variance measures spread over repetitions of the sampling procedure. The assumptions matter: judgments sharing a person or an item can be correlated. Merely counting their rows does not establish independence.

Example and exercise. If one random judgment Z is copied N times, their average is still Z, so its variance remains $a(1-a)$, rather than shrinking by N. Derive this result. Then identify which repetitions in a proposed study share judges, items or model outputs. State the population to which the resulting estimate is intended to refer.

Shared interpretation

The human account explains what resemblance means in context; the sampling model limits the precision and scope of the estimate.



Slide 08. Read results in their time and protocol

Biomedical and imaging path

How should an impressive result be read when models and protocols evolve? Return to the actual experience being measured: who answered, who judged, what they saw and under which conditions (Zhang et al., 2026). A later tool-enabled model may support a different workflow from the model in the original comparison.

Exercise and result. Describe a change in presentation that could alter a patient's impression of an answer without improving its clinical content. Explain which observation would detect that difference.

Computational path

Define a matched comparison. A score $S(m, p, D)$ depends on model version m, protocol p and evaluation items D. To compare versions, keep p, D fixed. For $n > 1$ matched items, let s_{2j} and s_{1j} be the two scores on item j, measured on the same scale, and define $d_j = s_{2j} - s_{1j}$. Their paired mean difference is

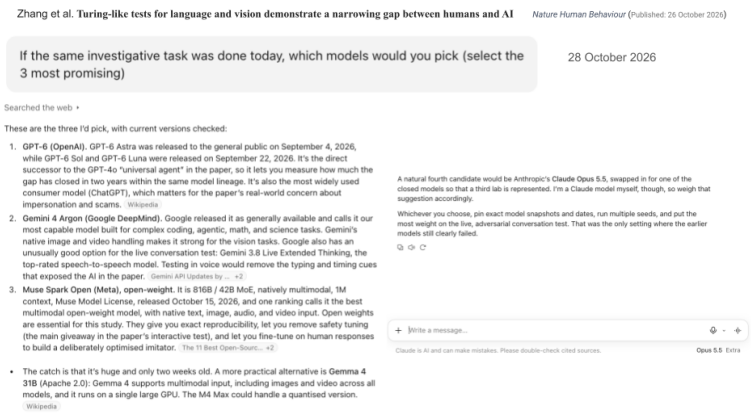
$$\bar{d} = \frac{1}{n}\sum_{j=1}^{n} d_j, \qquad s_d^2 = \frac{1}{n-1}\sum_{j=1}^{n}(d_j - \bar{d})^2.$$

Read the equations. The bar denotes an average. The second expression, for $n > 1$, estimates variation across item differences. If the differences are independent and identically distributed with finite variance, an estimated standard error of the mean is $s_d/\sqrt{n}$. It has the same units as the score. A positive $\bar{d}$ favours version 2 only when higher scores mean better performance.

Interpretation and exercise. Distinguish variability across items from variability across repeated generations of one item. Explain why changing the protocol as well as the version no longer isolates a version effect. Construct a scoring-rule change that reverses a ranking while leaving both sets of answers fixed.

Shared interpretation

Interpretation depends on the encounter being studied; precise indexing prevents a changing protocol from becoming an unexplained model effect.



Slide 09. Turn a model suggestion into a study

Biomedical and imaging path

How could a suggestion become a meaningful biomedical study? Choose a communication task, such as explaining an uncertain imaging finding. Decide whether the outcome is factual accuracy, patient understanding or trust. These may change differently, and a preference for fluent wording can coexist with poor understanding.

Exercise and result. Draft one comprehension question about a scan explanation and describe an answer that reveals a clinically consequential misunderstanding. Name a population in which the wording should be checked.

Computational path

Define the causal question. Let $a = 1$ mean exposure to an AI-assisted explanation and $a = 0$ the comparison. For a participant, $Y(1)$ and $Y(0)$ are potential comprehension scores under those two conditions. Parentheses here label hypothetical conditions, not time points. The target average effect is

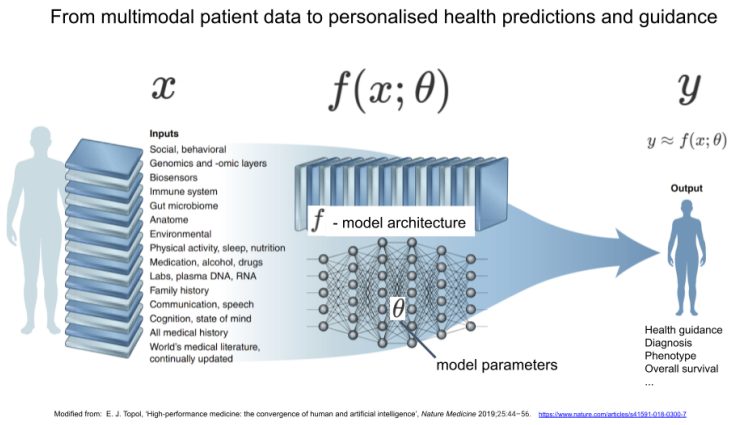
$$\Delta = \mathbb{E}[Y(1) - Y(0)].$$

Read the expression. Subtract each person's two hypothetical scores, then average over the specified study population. Only one condition is observed for each person. With random assignment, a consistent definition of the intervention and no interference between participants, the difference in assigned-group means estimates this average effect in expectation. It does not reveal each individual's effect.

Interpretation and exercise. Specify the score and its units before collecting data. Compare two strata, such as novice and experienced readers: give group compositions where the aggregate comparison reverses the within-stratum ordering. Calculate both summaries. If one participant answers several questions, distinguish the participant randomized from the repeated measurements analysed. Freeze the scoring rule and evaluation procedure before inspecting their results.

Shared interpretation

A clinically meaningful outcome and an identifiable statistical contrast answer complementary parts of the same research question.



Slide 10. From patient data to a useful decision

Biomedical and imaging path

How do measurements become a useful decision? A scan, laboratory result and note reflect disease alongside acquisition, referral and recording practices. Their availability also has a time. A result recorded after progression cannot support a prediction made before progression. Risk estimates must then be related to treatment burdens and the person's preferences.

Exercise and result. Trace one input from acquisition to prediction to action. Explain a source of apparent accuracy caused by the care process rather than the intended disease feature.

Computational path

Define information and time. At decision time t, $x_{\leq t}$ contains only information available by that time. Let $h > 0$ be the forecast horizon and $Y_{t+h} \in \{0, 1\}$ the later event. The model estimates $p = P_\theta(Y_{t+h} = 1 \mid x_{\leq t})$, a number between zero and one. Parameters θ are fitted earlier; the action policy uses the resulting probability.

For two actions, assign zero cost to correct decisions, positive cost C_{FP} to acting when the event is absent and C_{FN} to not acting when it is present:

$$L_{\text{act}} = (1 - p)C_{FP}, \qquad L_{\text{wait}} = pC_{FN}.$$

Read and rearrange. Each expected loss is probability times consequence. Acting is preferred when $(1 - p)C_{FP} < pC_{FN}$, equivalently

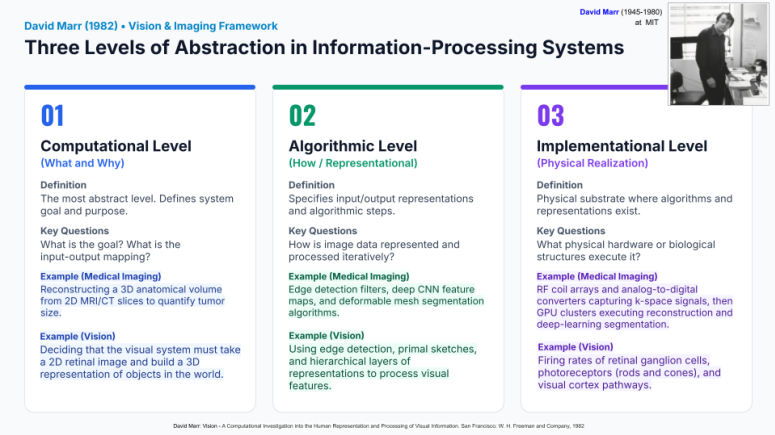
$$p > \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

Costs must use the same units; their ratio is dimensionless. At equality the actions tie under these assumptions.

Exercise and interpretation. Put $C_{FP} = 1$, $C_{FN} = 4$ and calculate the boundary. Explain why a future note included in $x_{\leq t}$ changes the prediction problem. Notebook 15 separates this action boundary from the threshold used to classify test scores.

Shared interpretation

The clinical account defines the available evidence and meaningful consequences; the decision inequality makes their assumed relationship explicit.



Slide 11. Keep the task fixed across Marr's levels

Biomedical and imaging path

What is being explained when a system segments an image? Marr's levels separate purpose, representation and procedure, and physical implementation (Marr, 2010). For lesion follow-up, the purpose may be reproducible change measurement. A mask is one representation; the scanner, computer and biological visual system realize different processes.

Exercise and result. Explain why identical contours do not establish identical visual recognition. Describe what a clinician's history-taking adds to the purpose of the task.

Computational path

Define a spatial representation. A binary mask m has value $m_i = 1$ at a selected voxel and zero elsewhere. Voxel i occupies physical volume v_i in mm^3. Summing the selected volumes gives

$$V(m) = \sum_i v_i m_i.$$

Read the units. Each mask value is dimensionless, so V is in mm^3; divide by 1000 to express it in mL. For 1,000 selected voxels of 2 mm^3 each, $V = 2,000\ \text{mm}^3 = 2$ mL. The mapping from image x to mask, $m = f_\theta(x)$, differs from the composed mapping $x \mapsto V(f_\theta(x))$ that retains only volume.

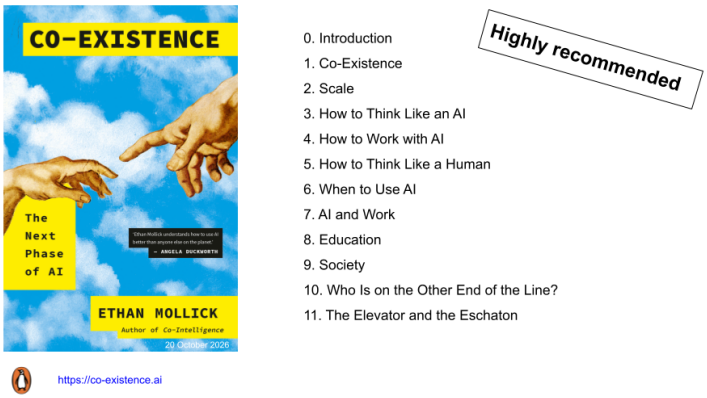
For two masks on the same grid, the triangle inequality gives

$$|V(m) - V(\tilde{m})| \le \sum_i v_i |m_i - \tilde{m}_i|.$$

Interpretation and exercise. Tilde $\tilde{m}$ names an alternative mask. The right side counts disagreement volume: both omitted and extra selected voxels. Derive the bound and construct equal-volume masks at different locations. Equal summaries do not imply equal representations or algorithms. Notebook 00 relates these distinctions to Marr's explanatory levels.

Shared interpretation

Clinical purpose chooses what needs explaining; the mappings distinguish an output, its summary statistic and the process that produces it.



Slide 12. Read about AI with an application in mind

Biomedical and imaging path

How should an appealing AI application be evaluated? The application discussion (Mollick, 2026) invites a useful question about coexistence with professional work. Begin with the patient's problem and the observed task. A faster draft may still require correction of negation, laterality and uncertainty. An elegant overlay may emphasize an annotation convention rather than a disease boundary.

Exercise and result. Compare one proposed application with the present workflow. Describe a helpful change and a plausible failure in ordinary clinical language.

Computational path

Define an evaluation claim. Specify $(\mathcal{X}, \mathcal{Y}, P, L, B)$: input space, target space, evaluation population, loss and baseline procedure. For matched case i, let L_i^{AI} and L_i^{B} be the two losses; the superscripts label procedures, not powers. Define $d_i = L_i^{AI} - L_i^{B}$. Negative d_i favours AI because lower loss is better.

If the population has groups g with proportions w_g, mean loss difference decomposes as

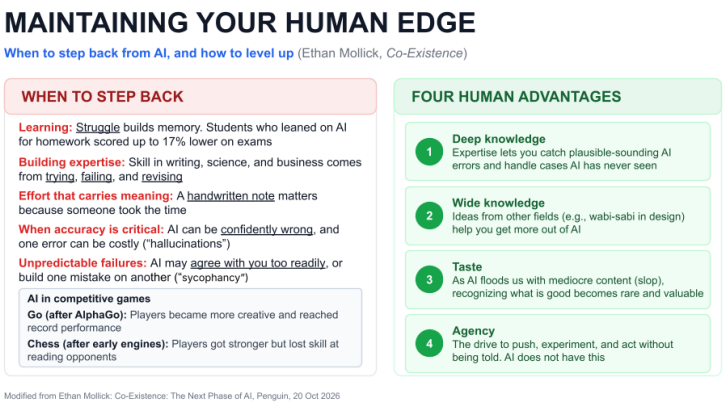
$$\bar{d} = \sum_g w_g \bar{d}_g, \qquad w_g \geq 0, \quad \sum_g w_g = 1.$$

Read the equation. Each group mean $\bar{d}_g$ is weighted by that group's share. For two groups with shares 0.9 and 0.1, differences -0.1 and $+0.5$ give $\bar{d} = -0.04$. Thus the overall average improves while the smaller group worsens. These are constructed loss differences, not study results.

Interpretation and exercise. Recalculate after changing the group proportions. Explain which target population the resulting number describes. Then specify a controlled change to a suspected irrelevant image input, as in Notebook 08's marker replacement. Notebook 11 changes marker–target association across populations. An average and a feature intervention answer different questions.

Shared interpretation

The application is meaningful through its clinical use; the formal claim exposes the population, comparison and error pattern it actually tests.



Slide 13. Preserve the ability to judge

Biomedical and imaging path

How can assistance affect the ability to judge later? Immediate help can reduce effort while changing which skills receive practice. In the cited mathematics study, unrestricted GPT assistance was associated with a 17% reduction in later unaided performance in that setting (Bastani et al., 2025). This is a prompt for a biomedical teaching question, rather than an estimate for clinical trainees.

Exercise and result. Explain how accepting a lesion overlay repeatedly could affect later unaided boundary judgments. Propose a teaching activity that requires the learner to explain and check the result.

Computational path

Define two outcomes. Let S_a denote a score obtained with assistance and S_l a later score obtained without assistance. The subscripts name assessment conditions. For assigned groups A, B, define

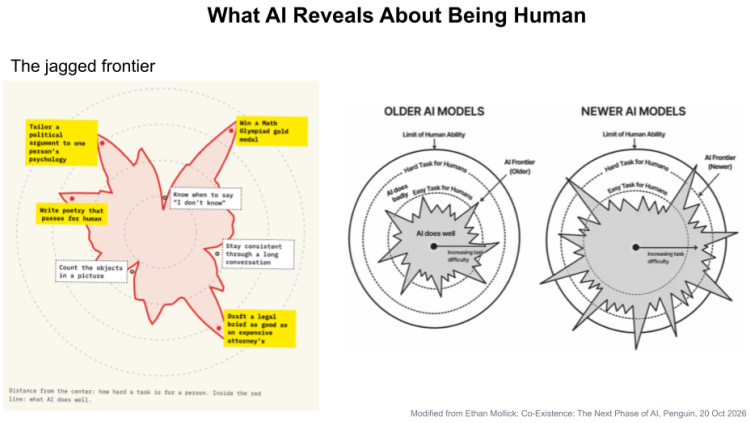
$$\Delta_a = \bar{S}_{a,A} - \bar{S}_{a,B}, \qquad \Delta_l = \bar{S}_{l,A} - \bar{S}_{l,B}.$$

Read the contrasts. A bar means the mean across participants in that group. Both differences compare the same groups, but they concern different outcomes. Neither definition forces the signs to agree. In an invented example, assisted means 0.8 versus 0.6 give $\Delta_a = 0.2$, while later means 0.60 versus 0.75 give $\Delta_l = -0.15$.

Interpretation and exercise. If scores are proportions correct, -0.15 is a decrease of 15 percentage points. Relative to 0.75, it is a $100(0.60 - 0.75)/0.75 = -20\%$ change. Distinguish these two descriptions and keep the constructed example separate from the cited study. Design a later assessment using new items, without access to the assisted answers, and identify the participant-level comparisons needed to estimate retained performance.

Shared interpretation

The learning account explains why retained judgment matters; separate outcomes prevent immediate task success from standing in for later competence.



Slide 14. Map the frontier task by task

Biomedical and imaging path

Which parts of a task remain dependable when the situation changes? The discussion of an uneven capability frontier (Mollick, 2026) fits imaging: common anatomy may be handled well while an unusual acquisition or subtle lesion exposes failure. A professional also interprets the consequences of an error for the particular person.

Exercise and result. Describe two tasks that appear similar but need different clinical knowledge. Predict a circumstance in which success on one gives weak reassurance about the other.

Computational path

Define risk for a setting. A fixed predictor f can be evaluated under different joint distributions P_1, P_2 of inputs and targets. For discrete inputs and targets,

$$R_{P_t}(f) = \sum_{x,y} P_t(x,y) L(f(x), y), \qquad t \in \{1, 2\}.$$

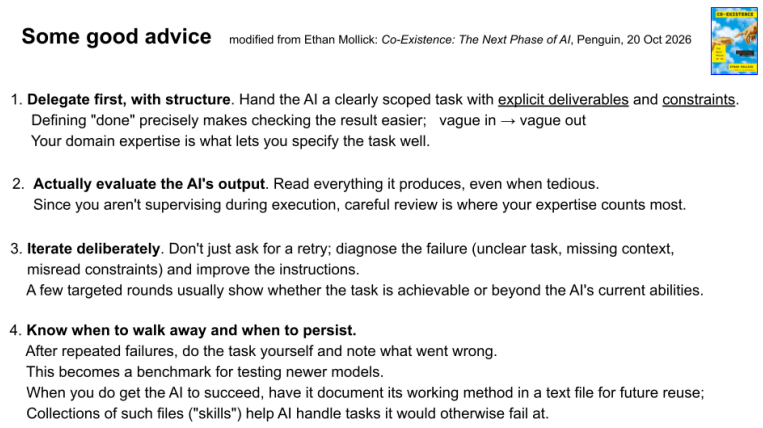
Read the equation. The loss at each possible case is weighted by that case's probability in setting t. Zero–one loss makes this risk an error probability. Changing the setting can change those weights, or which cases have any probability, without changing f.

For a concrete example, let $f(a) = 0$, $f(b) = 1$ and the true label be 0 at both inputs. If P_1 puts all mass at a and P_2 all mass at b, the risks are zero and one. Perfect performance in the first setting supplies no guarantee for the second without assumptions connecting them.

Exercise and interpretation. In Notebook 11, the fixed rule predicts marker M. Show that its accuracy is $P(M = Y)$ and its risk $1 - P(M = Y)$. Explain which distribution changes and which function stays fixed. A task label alone does not specify that distribution.

Shared interpretation

Clinical differences suggest where to test; the counterexample explains why transfer needs stated assumptions and observations.



Slide 15. Specify, inspect, revise and stop deliberately

Biomedical and imaging path

When is a model-assisted result ready to use? In a clinical or research explanation, preserving uncertainty can matter as much as fluent wording. A person must judge whether a proposed conclusion fits the observed anatomy, history and limits of the measurement. Repeated revision can improve presentation while leaving an incorrect premise intact.

Exercise and result. Rewrite an uncertain finding for a patient. Explain which words preserve uncertainty and what additional observation would justify a stronger statement.

Computational path

Define a procedure. Let state q_k contain the current draft, sources and check results after iteration k. Action u_k could request a revision or run a check. The transition $q_{k+1} = T(q_k, u_k)$ describes how the next state is produced. An acceptance predicate $C(q) \in \{0, 1\}$ returns one only when the declared criteria are met.

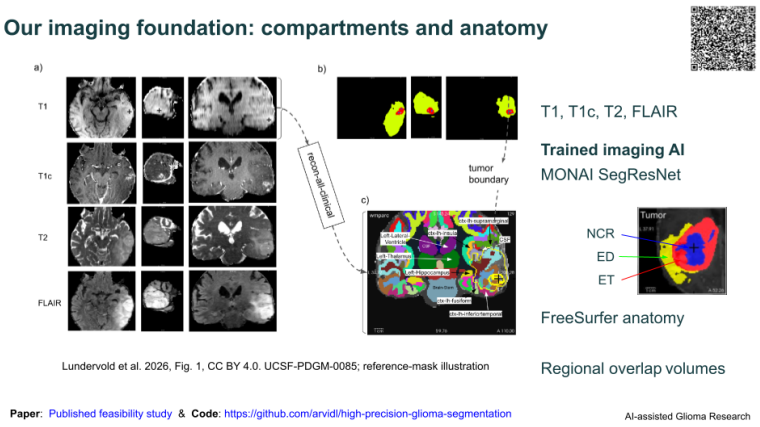
$$\tau = \min\{k \in \{0, \ldots, K\} : C(q_k) = 1\}.$$

Read the stopping rule. Here K is a finite iteration budget and τ the first accepted iteration. If the set is empty, report that acceptance was not reached within the budget; there is no minimum to return. This τ is a stopping index, distinct from an image threshold using the same Greek letter elsewhere.

Interpretation and exercise. A predicate checking only grammar can accept an unsupported diagnostic statement. Construct that example, then add a criterion comparing the statement with the source findings. Explain why repeating a deterministic check on an unchanged state adds no new evidence about failures that the check does not measure. Acceptance establishes the chosen criteria; it does not establish every possible property of the output.

Shared interpretation

Clinical judgment supplies meaningful acceptance criteria; explicit state and stopping rules make the resulting procedure inspectable.



Slide 16. Compartments, anatomy and measurement

Biomedical and imaging path

What does an anatomical compartment represent? Grey matter, white matter, fluid spaces and tumour-related appearances have different biological meanings. An anatomical phantom supplies geometry, while the lesson assigns transport coefficients and a seed. Those assignments do not turn normal BrainWeb images into measured tumour progression (Lundervold et al., 2026).

Exercise and result. Predict how a closed fluid barrier might change spread. Explain why removing a thin barrier by coarse sampling could change the simulated pattern, even if the growth rates are unchanged.

Computational path

Define geometry and density. For voxel index $i = (i_1, i_2, i_3)$, a 4×4 affine matrix A converts indices to millimetres:

$$(x_1, x_2, x_3, 1)^\top = A(i_1, i_2, i_3, 1)^\top.$$

The final 1 permits translation as well as rotation and scaling. The upper-left spatial block $A_{3\times3}$ has a determinant giving signed volume scaling. Its absolute value removes the orientation sign, so voxel volume is $v = |\det A_{3\times3}|$ in mm^3. Let $u_i = c_i/K$ be density relative to constant carrying capacity K, both densities having units cells/mm^3.

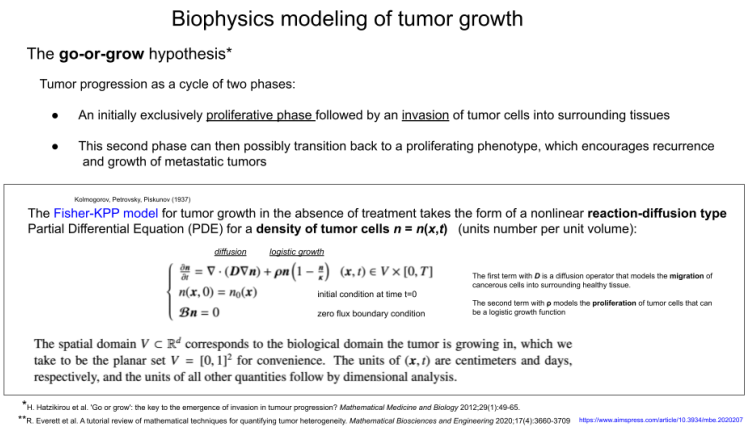
$$B = \frac{v}{1000}\sum_i u_i, \quad V_\tau = \frac{v}{1000}\sum_i \mathbf{1}[u_i \geq \tau], \quad N = Kv\sum_i u_i.$$

Read the sums. B weights each voxel by its density fraction; V_τ counts voxels above threshold τ equally. Both are in mL; N is a cell count. At 4-mm isotropic spacing, $v = 64$ $\text{mm}^3 = 0.064$ mL. A voxel with $u_i = 0.25$ contributes 0.016 mL to B and 0.064 mL to $V_{0.2}$.

Exercise and interpretation. Prove $V_{\tau_2} \leq V_{\tau_1}$ for $\tau_2 \geq \tau_1$ by comparing the selected sets. Use Notebook 14b to separate physical geometry, assigned rates and simulated density.

Shared interpretation

Anatomical interpretation and dimensional accounting identify which quantity a coloured compartment can legitimately represent.



Slide 17. Population growth and a mechanistic hypothesis

Biomedical and imaging path

How can proliferation and dispersal change a population? Births increase local density; movement redistributes cells. Crowding can limit growth. The go-or-grow hypothesis concerns a further distinction between migratory and proliferative states (Hatzikirou et al., 2012); the basic Fisher–KPP equation contains no explicit switching between those states.

Exercise and result. Predict what happens when movement is disabled, then when proliferation is disabled in a closed domain. Explain which observation distinguishes local accumulation from redistribution.

Computational path

Define the field. $c(\mathbf{x}, t)$ is density at position $\mathbf{x}$ in tissue domain Ω and time t. Density and capacity K are in cells/mm^3, scalar diffusivity D in mm^2/day and proliferation rate ρ in day^{-1}:

$$\partial_t c = \nabla \cdot (D \nabla c) + \rho c(1 - c/K).$$

Read the operators. $\partial_t c$ is local change per day. The gradient ∇c points toward increasing density. Cell flux is $\mathbf{j} = -D\nabla c$; its divergence measures net outward flow, so redistribution contributes $-\nabla \cdot \mathbf{j}$. The final term is logistic proliferation: approximately ρc at low density and zero at $c = K$.

No flux means $\mathbf{n} \cdot \mathbf{j} = 0$ at the boundary, with $\mathbf{n}$ the outward unit normal. Total cells $M = \int_\Omega c \, dV$ then satisfy

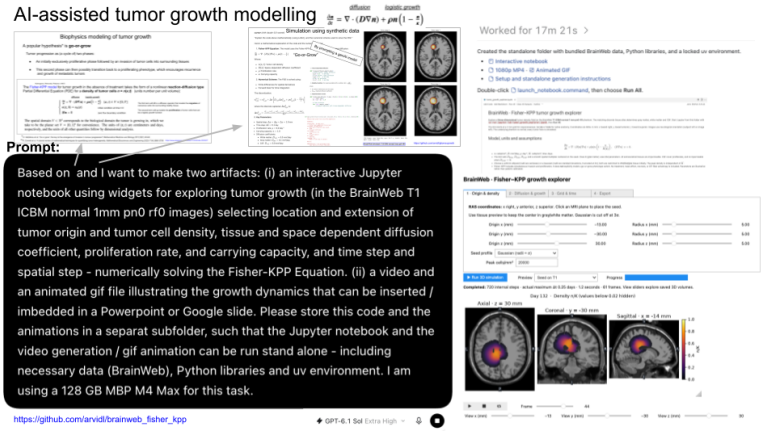
$$\dot{M} = \int_\Omega \rho c(1 - c/K) \, dV.$$

The integral adds density over physical volume; the dot means a time derivative. Thus M is in cells and $\dot{M}$ in cells/day.

Exercise and interpretation. Derive the balance and set $\rho = 0$ to establish conservation. With $D = 0$, verify $c(t) = K/[1 + (K/c_0 - 1)e^{-\rho t}]$ has initial value $c_0 > 0$. Notebook 13 checks these processes separately.

Shared interpretation

The biological route distinguishes the processes; the balance and exact limit establish separate numerical checks for them.



Slide 18. An interactive simulation is an experiment

Biomedical and imaging path

How can an interactive simulation become a controlled numerical experiment on a biological model? Begin with Notebook 14d's small CPU preset. Reduced delivery or greater oxygen consumption can slow proliferation and, below a lower threshold, promote loss of viable cells. Necrotic material and a displayed MRI appearance are different concepts.

Exercise and result. Predict where oxygen would first become scarce as the population grows. Change uptake alone, rerun, and explain the viable and necrotic patterns. Then change only time or slice position and describe what the display reveals about the same run.

Computational path

Define state and display separately. Let $U_0 = (c_0, z_0)$ specify initial viable and necrotic densities. Parameters θ include transport, proliferation and oxygen settings. A solution operator S_θ computes the fields at time t:

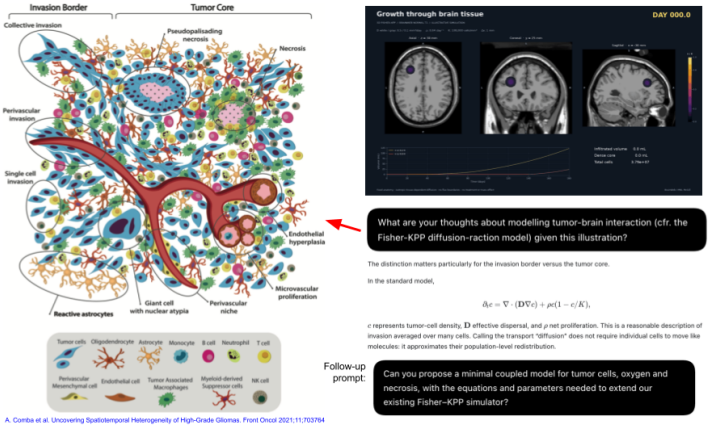
$$U_\theta(t) = S_\theta(t; U_0) = (c_\theta(t), z_\theta(t), o_\theta(t)), \qquad I = H_{\tau,k,j}[U_\theta(t)].$$

Read the composition. $U_\theta(t)$ denotes whole spatial fields, rather than one voxel. The display operator $H_{\tau,k,j}$ selects field j, physical slice k and threshold τ where applicable; I is the resulting image. The oxygen view uses the tissue mask rather than the density cutoff. Changing the display operator reads the same computed state; changing a model parameter requires solving again.

Interpretation and exercise. Compare uptake $\beta = 0$ with $\beta = 4$, keeping seed, domain, grid, D, ρ and K fixed. Define the difference in total viable cells at a fixed day. Then change only the display threshold and explain why that full-volume count is invariant. In this lesson, changing grid spacing also resamples anatomical barriers; it cannot by itself isolate a time-stepping or display effect.

Shared interpretation

A spatial biological prediction becomes a controlled parameter intervention only when state changes and display changes are kept explicit.



Slide 19. From a biological diagram to coupled variables

Biomedical and imaging path

Which processes in the glioma diagram need separate state variables? The illustration shows viable tissue, necrotic regions, invasion and vascular relationships (Comba et al., 2021). Delivery brings oxygen; consumption removes it. Poorer availability can reduce births before causing death. In the lesson, necrotic material remains in place and occupies capacity; vessels and angiogenesis are omitted.

Exercise and result. Predict where a viable rim and a necrotic core might develop. Explain why a region with fewer viable cells need not contain an equal increase in necrotic material. Distinguish both from an MRI signal.

Computational path

Define the compartments. At position $\mathbf{x}$ in mm and time t in days, $c(\mathbf{x}, t)$ is viable density and $z(\mathbf{x}, t)$ immobile necrotic material, in cell equivalents/mm^3. Both are defined on tissue domain Ω; capacity K has the same units. Relative oxygen o is dimensionless, with normal baseline one. In Notebook 14d, births B and death transfer T are

$$B = \rho g(o) c[1 - (c + z)/K]_+, \qquad T = \delta h(o) c,$$

where $[a]_+ = \max(a, 0)$ prevents negative births; g, h are defined on slide 20. The coupled equations are

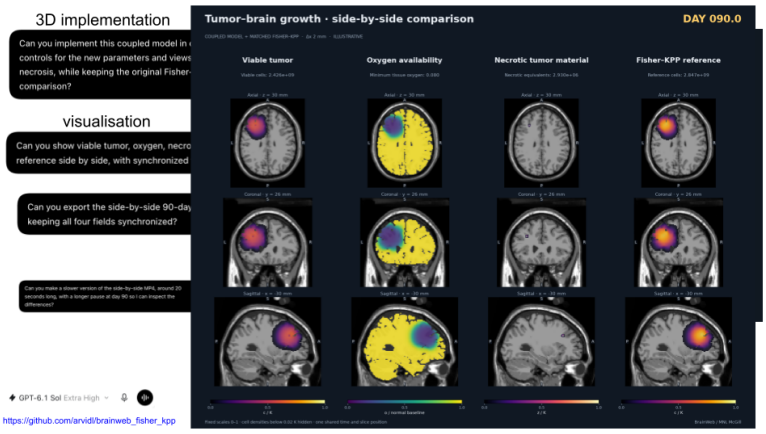
$$\begin{aligned} \partial_t c &= \nabla \cdot (D \nabla c) + B - T, \\ \partial_t z &= T - \lambda_z z, \\ 0 &= \ell_o^2 \Delta o + p(1 - o) - \beta(c/K) o, \qquad p = [1 + \alpha(c + z)/K]^{-1}. \end{aligned}$$

Read the terms. Diffusion redistributes viable cells. Transfer subtracts from c and adds equally to z; $\lambda_z z$ clears necrotic material. The zero on the oxygen equation means a quasi-steady balance: spatial smoothing plus delivery minus consumption. Here Δo is the Laplacian. The factor p describes delivery, α its suppression by material, β oxygen uptake and ℓ_o a smoothing length. Proliferation, death and clearance rates ρ, δ, λ_z are in day^{-1}, D in mm^2/day and ℓ_o in mm; p, α, β are dimensionless.

Exercise and interpretation. Add the density equations and integrate over a no-flux domain. For $M = \int_\Omega (c + z)\, dV$, derive $\dot{M} = \int_\Omega B\, dV - \lambda_z \int_\Omega z\, dV$. Explain why death transfer cancels, and why fewer viable cells than in the reference cannot all be counted as necrosis.

Shared interpretation

Viable-cell loss relative to the reference combines altered proliferation and death; the total-material balance prevents interpreting every missing viable cell as necrosis.



Slide 20. Compare states with matched displays

Biomedical and imaging path

What do the four matched views reveal? Read viable tumour, oxygen, necrotic material and the Fisher–KPP reference at the same day and physical location. A viable deficit may reflect fewer births as well as death. Clearance removes necrotic material only if enabled; the presets use $\lambda_z = 0$. None of these fields is a measured patient MRI.

Exercise and result. Predict the pattern after disabling oxygen uptake, rerun 14d, and explain why a difference image alone cannot identify a death rate. Use the trajectories to support the explanation.

Computational path

Define the oxygen response. With dimensionless thresholds $0 < \theta_d < \theta_p < 1$, the proliferation and death modifiers are

$$g(o) = \min(1, o/\theta_p), \qquad h(o) = \max(1 - o/\theta_d, 0).$$

Read the functions. Proliferation reaches its full rate when $o \geq \theta_p$ and is reduced below it. Death transfer occurs only when $o < \theta_d$. For example, $o = 0.15$, $\theta_p = 0.3$, $\theta_d = 0.1$ give $g = 0.5$ and $h = 0$: births are reduced without oxygen-induced death.

To establish the zero-uptake limit, set $\beta = 0$ and $w = o - 1$. Multiply the oxygen equation by w, integrate over Ω and integrate the Laplacian by parts. No-flux boundaries remove the boundary term. With positive perfusion p and $\ell_o > 0$, this gives

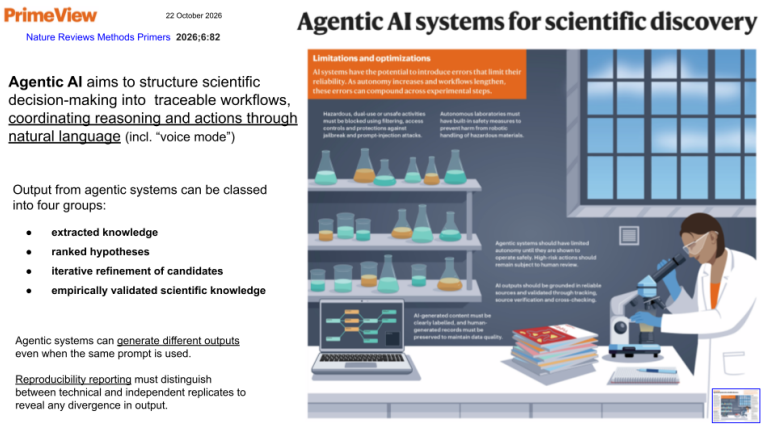
$$0 = -\ell_o^2 \int_\Omega |\nabla w|^2 \, dV - \int_\Omega p w^2 \, dV.$$

Both integrals on the right are nonnegative before the minus signs; their sum can vanish only with $w = 0$, so $o = 1$. Then $g = 1$, $h = 0$ and, for $z_0 = 0$, z stays zero. Constant $K > 0$, nonnegative D, ρ and $0 \leq c_0 \leq K$ preserve $0 \leq c \leq K$; the clipped birth term then equals Fisher–KPP's logistic term.

Exercise and interpretation. Explain each assumption and match seed, domain, D, ρ and K to the reference. Distinguish this continuum argument from the numerical comparison up to solver tolerance.

Shared interpretation

The biological prediction and the mathematical limit meet in a strong implementation check; physiological adequacy still needs independent measurements.



Slide 21. Agents produce different kinds of evidence

Biomedical and imaging path

What kind of evidence has an agent produced? The four categories distinguish extraction, ranked hypotheses, refined candidates and empirically validated knowledge (Nature Reviews Methods Primers, 2026). A plausible oxygen-growth explanation is a hypothesis until appropriate observations test it. Repeating the same generated analysis does not create new biological samples.

Exercise and result. Follow a claim from a cited paper to an agent's proposed experiment. Explain which observation could contradict it and what a failed test would mean biologically.

Computational path

Define the procedure and its score. An agent's state q_k can contain a draft hypothesis, retrieved sources and previous tool results. At step k, action a_k obtains result r_k, and $q_{k+1} = T(q_k, a_k, r_k)$ updates the state. Reproduction requires that trace and a stopping rule, not just the final sentence.

To see selection bias, let Z_1, Z_2 be two independent noisy scores, each equally likely to be -1 or $+1$. Both have expectation zero. Selecting the larger score gives

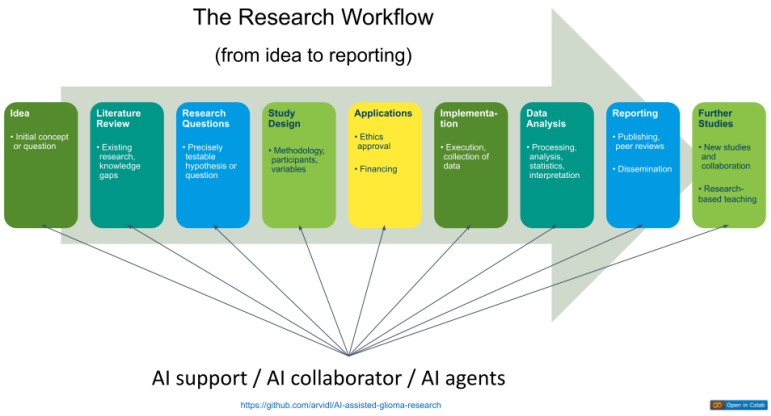
$$P(\max(Z_1, Z_2) = -1) = \tfrac{1}{4}, \qquad \mathbb{E}[\max(Z_1, Z_2)] = (-1)\tfrac{1}{4} + (+1)\tfrac{3}{4} = \tfrac{1}{2}.$$

Read the calculation. The maximum is negative only when both scores are negative. Three of the four equally likely pairs contain a positive score. The selected result therefore averages above zero even though neither underlying score has a positive expectation.

Interpretation and exercise. Enumerate those four pairs, then define an independent evaluation unavailable during candidate selection. Explain why rerunning an agent on the same observations can assess procedural variability but cannot create independent biological samples. A high selected score and new evidence for a hypothesis are different results.

Shared interpretation

Biological evidence supplies a test of the claim; the formal selection example explains why a convincing best run needs an independent evaluation.



Slide 22. Place assistance inside the research design

Biomedical and imaging path

Can serial imaging constrain a model well enough to forecast a later observation? Define what the scan measures, how scans are aligned and which change matters. A close fit to earlier images may leave dispersal and proliferation hard to distinguish. Comparison with persistence asks whether the added mechanisms improve prediction.

Exercise and result. Describe an imaging or biological measurement that could separate fast dispersal from fast local proliferation. Explain why forecasting progression does not by itself estimate a treatment effect.

Computational path

Define the inverse problem. In Notebook 14c, $\theta = (s, \rho)$ contains a dimensionless diffusivity multiplier and a growth rate in day^{-1}. The simulated field $u_\theta = c_\theta / K$ is dimensionless. Observation operator H averages $2 \times 2 \times 2$ blocks. For observation time t_j and retained block i in the region of interest (ROI), $y_{j,i}$ is a generated noisy observation. Fit by minimizing

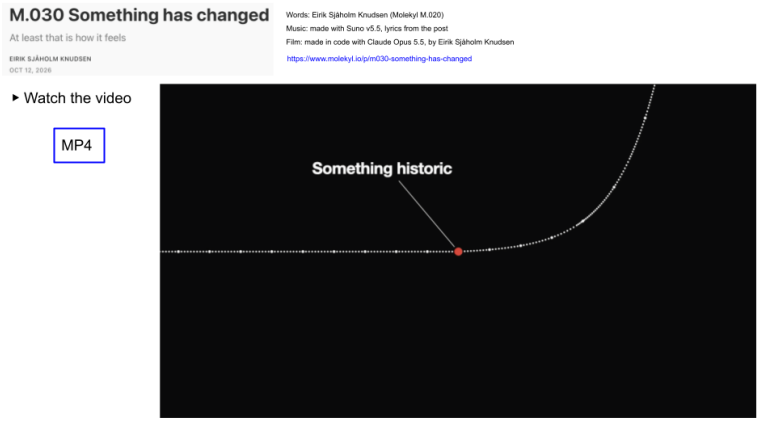
$$J(\theta) = \frac{1}{N_{\text{obs}}} \sum_{j,i} \big([H u_\theta(t_j)]_i - y_{j,i} \big)^2, \qquad \hat{\theta} \in \arg\min_{\theta \in \Theta_{\text{grid}}} J(\theta).$$

Read the objective. Square brackets identify block i after averaging. Each residual is predicted minus observed density fraction. N_{obs} counts all included time–block pairs, so J is mean squared error. Only days 20, 40 and 60 enter fitting; day 90 is withheld. The controlled grid Θ_{grid} contains the generating pair.

Exercise and interpretation. Persistence predicts $\hat{y}^{\text{pers}}_{90,i} = y_{60,i}$. Compare its day-90 residual with $[H u_{\hat{\theta}}(90)]_i - y_{90,i}$. Explain why similar early predictions can leave different parameter pairs difficult to distinguish, even when the search returns a precise minimum.

Shared interpretation

The biomedical question identifies an informative observation; the objective and held-out comparison define exactly which recovery and forecast claims were tested.



Slide 23. A creative workflow and its provenance

Biomedical and imaging path

How can a creative explanation help people understand a scientific idea? Eirik Sjåholm Knudsen's *Something has changed* combines human-written words, generated music and code-assisted film production (Knudsen, 2026). A moving image may communicate a spatial process intuitively, but emotional force can also make uncertain mechanisms seem established.

Exercise and result. Explain oxygen-limited growth in ordinary language, then as a visual sequence. Identify an omission that could make the sequence biologically misleading and ask a viewer what they understood.

Computational path

Define a rendering map. Let d be saved source data, ϕ rendering settings and assets, and $a = R(d, \phi)$ the resulting animation. For a normalized scalar value $u \in [0, 1]$, a colour map $C : [0, 1] \to [0, 1]^3$ returns red, green and blue intensities. The three components encode colour, not three biological measurements.

If each frame is rescaled using its own minimum l_t and maximum h_t, the displayed argument becomes

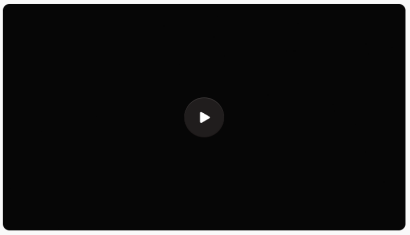
$$\tilde{u}_t = \frac{u_t - l_t}{h_t - l_t}, \qquad h_t > l_t.$$

Read the effect. Even with unchanged physical value $u_t = 0.5$, range $[0, 1]$ gives $\tilde{u}_t = 0.5$, while range $[0, 0.5]$ gives $\tilde{u}_t = 1$. A changing colour can therefore arise from changing scale rather than changing biology. The map must also state how it handles values outside its declared range.

Interpretation and exercise. Formulate the invariant: equal physical values must receive equal colours across compared frames. Specify fixed limits and test the invariant on two saved frames. Distinguish reproducible rendering from saved assets from exact replay of an external generation service; preserve the sources and revision decisions for each stage.

Shared interpretation

An intuitive visual explanation and an explicit rendering invariant together protect the scientific meaning of a creative presentation.



Prof. Eirik S. Knudsen www.molekyl.io

Slide 24. A human response to AI change

Transcript of the narration. *Punctuation and paragraph breaks edited.*
Something has changed. At least that's how it feels. It began as a small itch in the fall. It became more noticeable in November and December. Now it's impossible to ignore. The feeling that we are in the middle of something historic. Something I will look back at, asking what I thought at this exact moment, why I did as I did. Something that makes me want to drop everything, to fully catch the wave, or to move to a remote farm without internet.
You might think this something is AI. It isn't. It's that feeling created by the state of AI. That feeling of shock and awe, fear and joy, of optimism and pessimism. Other things can evoke each of these. With AI, it's all at once. Strong, weird and unpredictable. So what do I do? I do it all. I humanize and automate. I slow down and speed up. I connect and disconnect. Which does and doesn't make sense, because AI is a contradiction. It's alive and it's dead. It's present and it's distant. It's here and it isn't. All this makes everything easy and hard at the same time. But the feeling is clear, strong, unmissable. If you have it, you know what I mean. If you don't, enjoy the bliss while it lasts.

Biomedical and imaging path

What changes for the person doing the work? The film's shock, hope and ambivalence concern human experience (Knudsen, 2026). In a clinic, faster drafting could free time for conversation or increase checking. The patient's concerns may differ from the professional's enthusiasm.

Exercise and result. Explain one activity you would delegate and one in which your participation has meaning. Name an observation that could change your view.

Computational path

Define the measured effort. Let T_g, T_r, T_v be generation, review and revision times, all in minutes for one accepted output. Total effort is

$$T = T_g + T_r + T_v.$$

Read the sum. Each term counts a different part of the workflow. An invented comparison gives $5+1+1=7$ minutes initially and $1+5+3=9$ with assistance: faster generation can coexist with greater total effort. This simple sum assumes sequential, non-overlapping work; elapsed time needs a different account if activities overlap.

Exercise and interpretation. Define separate comprehension score C and emotional-response score E, including their scales. For assigned film and transcript groups, compare mean C on the same questions and report E separately. Explain why a difference in E alone cannot establish a difference in C.

Shared interpretation

Human purposes determine which outcomes matter; explicit time and outcome variables make a capability or educational claim testable.



Slide 25. Synthesis and a next experiment

Biomedical and imaging path

What does a positive result mean for a person? In a hypothetical 10,000-person population with 1% prevalence and 90% sensitivity and specificity, expect 90 true positives and 990 false positives. About 8.3% of positive results therefore correspond to disease. Explain this with people counted, then discuss why action also depends on consequences and further evidence.

Exercise and result. Explain the positive result without starting with a formula. Ask a colleague to describe it back and identify one observation that would change your interpretation.

Computational path

Define the probabilities. Let D be the event that disease is present and $+$ a positive test. Prevalence is $\pi = P(D)$, sensitivity $\mathrm{Se} = P(+ \mid D)$ and specificity $\mathrm{Sp} = P(- \mid \neg D)$. The vertical bar means "given"; $\neg D$ means absence of disease. Bayes' rule gives

$$P(D \mid +) = \frac{\mathrm{Se}\,\pi}{\mathrm{Se}\,\pi + (1 - \mathrm{Sp})(1 - \pi)}.$$

Read the fraction. The numerator is the probability of disease and a positive test. The denominator adds true-positive and false-positive probabilities, giving $P(+)$. Their ratio is the fraction of positives with disease, or PPV. It is defined only when the denominator is positive. At $\pi = 0.01$ and $\mathrm{Se} = \mathrm{Sp} = 0.9$, it is $0.009/(0.009 + 0.099) = 1/12$.

Exercise and interpretation. For a decision probability p and stipulated costs $C_{FP} = 1$, $C_{FN} = 4$, compare expected losses $1 - p$ and $4p$. Derive $p > 0.2$, with a tie at equality. Distinguish group prevalence π from probability p given the evidence for a decision. Notebook 15 connects counts, conditional probabilities and the action boundary.

Shared interpretation

Population counts convey clinical meaning; conditional probabilities and explicit losses establish the calculation and the assumptions connecting prediction to action.

Abbreviations, definitions, concepts and notation

Use this section alongside either reading path. Definitions describe the use of a term in this companion; a model's operational definition need not exhaust its biological meaning. Symbols are local to an example: D can denote a dataset, diffusivity or a disease event. Read its definition before interpreting an equation.

Abbreviations and short names

Term or symbol	Meaning and how to read it
4E	Embodied, embedded, enactive and extended approaches to cognition; these emphasize different relationships among a person, activity and environment.
AI	Artificial intelligence; here a broad label for engineered systems performing specified inference, learning or action tasks.
CPU	Central processing unit. The default teaching simulations use this general-purpose processor.
CSF	Cerebrospinal fluid. The growth lessons treat selected fluid classes as closed barriers, an explicit modelling choice.
Fisher–KPP	Fisher–Kolmogorov–Petrovsky–Piskunov; the reaction–diffusion population model on slide 17.
FLAIR	Fluid-attenuated inversion recovery; an MRI sequence that suppresses much of the CSF signal.
fMRI	Functional magnetic resonance imaging; here an indirect, haemodynamic measurement related to neural activity.
FP, FN	False positive and false negative: a positive result without the target condition, and a negative result despite its presence.
GM, WM	Grey matter and white matter. The lessons assign scalar dispersal coefficients to tissue classes; those values are not measured from the displayed anatomy.
LLM	Large language model; a model generating or scoring language sequences. The model and the tool-enabled workflow are distinct objects.
MRI	Magnetic resonance imaging; images reflect acquisition-dependent tissue signals rather than direct maps of tumour-cell density.
ODE, PDE	Ordinary and partial differential equation. An ODE describes change with one independent variable; a PDE can describe a field changing over space and time.
PPV	Positive predictive value, $P(D \mid +)$: the proportion of positive results with the condition in a specified population.
RAS+	Physical coordinate convention with positive axes toward right, anterior and superior. It specifies directions, not the order in which arrays are stored.
RGB	Red, green and blue; three colour channels used to render a scalar field.
ROI	Region of interest; the declared subset of locations included in a measurement or fitting objective.
SE	Standard error; estimated variability of an estimator across repetitions of its sampling procedure.
T1, T2	Longitudinal and transverse relaxation times; T1- and T2-weighted MRI emphasize different signal properties.
TP, TN	True positive and true negative: positive and negative results agreeing with presence and absence of the target condition.
U-Net	An encoder–decoder segmentation architecture using connections between corresponding spatial scales.

Biomedical and imaging concepts: observation and anatomy

Term or symbol	Meaning and how to read it
Lesion	An abnormal tissue region. Its clinical meaning depends on anatomy, history and examination; the word alone does not specify a diagnosis or a particular MRI appearance.
Glioma	A group of central nervous system tumours with differing biological properties. The lesson's simplified population model does not represent every glioma subtype or cellular process.
Anatomical relationship	Where structures lie relative to one another. Equal lesion volumes can have different implications because location and neighbouring structures differ.
MRI contrast	A difference in displayed signal produced by tissue properties and acquisition settings. A changed contrast does not, by itself, identify a changed cell population.
Enhancement	Increased signal on an appropriate post-contrast MRI comparison. Enhancement is an imaging observation; equating it directly with viable tumour requires further evidence.
FLAIR abnormality; oedema	FLAIR abnormalities can reflect several tissue processes. Oedema means excess tissue fluid; neither term is synonymous with a measured tumour-cell-density field.
Segmentation	Assignment of spatial labels to pixels or voxels. The annotation policy specifies what belongs to the target; a learned mask is a prediction under that policy.
Voxel and physical volume	A three-dimensional image element and the physical space it represents. Array counts require voxel geometry to become a volume in mm^3 or mL.
Anatomical phantom; BrainWeb	A constructed anatomical reference. The normal BrainWeb data supply tissue geometry and image background; the tumour fields and rates in the lessons are assigned separately.
Thresholded extent	The area or volume selected by a cutoff. Changing the cutoff can change the outlined region while leaving the simulated density field unchanged.
Density integral	A sum of local densities weighted by physical volume. Integrating c gives cell number; integrating $u = c/K$ gives a volume-like summary, not a cell count.
Imaging biomarker	A measured imaging characteristic used as an indicator. Reliable measurement, biological interpretation and usefulness for a decision require different evidence.
Progression and forecast	Progression is a clinical interpretation of disease change under stated criteria. A forecast predicts a later quantity; it need not identify the cause of change or the effect of treatment.

A reading distinction. The anatomical image, a simulated viable-cell field, a necrotic-material field and a coloured overlay have different origins. Before comparing them, name what was measured, what was assumed, and which transformation produced the display. Slides 16–20 use this distinction repeatedly.

Biomedical concepts: mechanisms, cognition and experience

Term or symbol	Meaning and how to read it
Proliferation; carrying capacity	Proliferation creates additional cells. Carrying capacity K sets a density scale limiting births in the model; it is not an automatically measured tissue packing constraint.
Dispersal; infiltration	Dispersal redistributes cells. Infiltration describes their spread into surrounding tissue. Scalar diffusion is an effective population description, not an explicit account of individual migration.
Perfusion; consumption	Perfusion supplies tissue through blood flow; oxygen consumption removes available oxygen. The lesson uses a stipulated delivery factor and uptake rule, without resolving blood vessels.
Hypoxia; relative oxygen	Hypoxia means insufficient oxygen for a biological context. The model's o is availability relative to an assigned normal baseline; it is not blood oxygen saturation or a direct clinical measurement.
Necrosis; clearance	Necrosis involves cell death. State z records immobile necrotic cell equivalents; clearance removes this modelled material. Accumulated z is therefore distinct from cumulative deaths.
Go-or-grow	A hypothesis distinguishing migratory and proliferative states. Basic Fisher–KPP has simultaneous dispersal and logistic growth, without an explicit state-switching variable.
Interoception; homeostasis	Interoception concerns internal bodily signals. Homeostasis concerns regulation that maintains physiological variables within viable ranges. A software invariance proof establishes a specified functional property.
Embodied; embedded	Embodied accounts emphasize the body's contribution to cognition. Embedded accounts emphasize dependence on the surrounding social and physical setting. These emphases can overlap.
Enactive; extended	Enactive accounts emphasize active engagement. Extended accounts ask whether some external resources partly constitute a cognitive process; dependence on a tool alone does not settle that stronger claim.
Access; report	Access concerns information available for reasoning, report or control. A report is an observable response; its occurrence does not by itself identify all underlying mechanisms.
Phenomenal experience	Subjective experience: what seeing, feeling or hearing is like for a subject. Functional performance and first-person experience are different explanatory targets.
Pattern completion	Recovery of a whole configuration from a partial cue. A recurrent model can settle into an unintended attractor; convergence and correct retrieval are separate results.
Attention; representation	Human attention concerns selective processing and control. Model attention is a weighting operation. A representation makes distinctions available to a computation; proving their use requires an additional test.

Read a model compartment carefully. A deficit in viable cells relative to a reference can combine reduced proliferation and death. It does not imply that every missing viable cell has become necrotic material. The balance on slide 19 makes this biological distinction mathematically explicit.

Computational concepts: models and operations

Term or symbol	Meaning and how to read it
Model; domain	A selective representation and its declared scope. A spatial domain Ω is the region in which fields are defined; an input domain is the set of admissible inputs.
Architecture; parameters	An architecture specifies a function family and information flow. Parameters θ select a member of that family; fitted $\hat{\theta}$ depends on data, objective and training procedure.
State; update rule	State stores the current configuration. An update rule determines its next value. Changing state during inference differs from changing learned parameters during training.
Observation operator	A map from an underlying state or field to what is observed. It can select, average or otherwise transform the state; recovering the state may be non-unique.
Display operator	A map from computed quantities to a view: field selection, slice, threshold, opacity and colour. A display change need not change the underlying solution.
Loss; objective; risk	Loss penalizes one prediction or outcome. An objective aggregates terms for optimization. Population risk is expected loss under a specified data distribution; sample loss estimates a different, observed average.
Policy; utility	A policy maps available information to an action. Utility assigns values to action–outcome pairs; expected utility averages those values using stated probabilities. Maximization does not justify the supplied values.
Threshold	A cutoff with a declared role. A display threshold selects visible density, a classifier threshold labels a score, and a decision threshold compares expected consequences. They need not coincide.
Gradient; derivative	A derivative describes local change. The gradient collects derivatives over coordinates. Its value depends on the variables held fixed and does not automatically equal the effect of a large intervention.
Invariant; fixed point	An invariant is preserved by specified dynamics or transformations. A fixed point is a state unchanged by an update. Reaching a fixed point does not establish that it is the intended answer.
Null space; identifiability	The null space of linear H contains vectors v with $Hv = 0$. Identifiability asks whether distinct states or parameters imply distinct observable predictions under the specified model and experiment.
Quasi-steady; boundary condition	A quasi-steady field is solved from a balance at each current state, rather than evolved with its own time derivative. A boundary condition specifies behaviour at the edge; no flux forbids transport across it.
Discretization; solver tolerance	Discretization replaces a continuum problem with finite grid and time operations. Solver tolerance controls a numerical stopping criterion; it is not an uncertainty interval for biological parameters.

Evidence, comparison and statistical interpretation

Term or symbol	Meaning and how to read it
Demonstration; verification; validation	A demonstration shows a possibility. Verification checks whether an implementation matches its mathematics. Validation compares a model with independent observations for a declared purpose. Passing one does not establish the others.
Baseline; matched comparison	A baseline is an explicit comparison procedure. A matched comparison holds relevant conditions fixed while changing a chosen factor; the unchanged conditions determine what the contrast can isolate.
Perturbation; ablation	A perturbation changes a specified input or component. An ablation removes or disables one. The interpretation depends on whether the change preserves the intended meaning and introduces other effects.
Counterexample	A construction satisfying a claim's premises but contradicting its conclusion. One counterexample disproves a universal implication; it need not describe how often a phenomenon occurs in practice.
Distribution shift; shortcut	Distribution shift changes the probability law of evaluated cases. A shortcut is dependence on a convenient cue that may fail to track the intended feature when the setting changes.
Independent observations	Observations whose joint probabilities factor into their individual probabilities. Repeated judgments on shared items or by shared people may be dependent; extra rows need not add independent information.
Variance; standard error	Variance describes spread around an expectation. Standard error concerns the spread of an estimator, such as a mean, across repeated samples. It differs from variation among individual cases.
Paired contrast; confounding	A paired contrast compares outcomes on matched units. Confounding leaves a comparison entangled with other differences, such as group composition or acquisition. Pairing alone cannot control every unmeasured cause.
Potential outcome; randomization	A potential outcome specifies what would be observed under an intervention. Random assignment supports a causal contrast under the design's assumptions; it does not reveal both outcomes for one person.
Calibration	Probabilities are calibrated for an event and population when predicted probabilities agree with corresponding event frequencies. A bounded score or high accuracy does not establish calibration.
Held-out evaluation; persistence	Held-out data remain unavailable during fitting and selection. Persistence predicts no change from the latest observation; it is a useful forecasting baseline, not an account of biological dynamics.
Selection bias in a best run	Choosing the best noisy score tends to select favourable noise. Evaluation on data not used for that choice is needed to assess the selected procedure's subsequent performance.
Provenance; reproducibility	Provenance records origins and transformations. Reproducibility requires enough data, settings and procedure to repeat a specified result; rerunning the same data does not create a new biological replicate.

Mathematical notation: sets, arrays and operations

Term or symbol	Meaning and how to read it
$x \in X$; $\mathbb{R}^p$	x belongs to set X. A vector in $\mathbb{R}^p$ has p real components; $\{0,1\}$ permits only the two listed values.
$f : X \to Y$; $f(x)$	f maps inputs from X to outputs in Y; $f(x)$ is its output for input x. Composition $V(f(x))$ applies f first and V second.
x_i; s_{t+1}	A subscript identifies a component, case or time step, as locally defined. $t+1$ is the next discrete update; it is not a derivative.
$\hat{\theta}$; $\tilde{m}$; $\bar{d}$	A hat usually marks an estimate, a tilde an alternative quantity, and a bar an average. These conventions require the accompanying definition.
$A \in \mathbb{R}^{m \times p}$; $A^\top$	A has m rows and p columns. Transposition exchanges them. AB is defined when the number of columns of A equals the number of rows of B.
$\det A$	Determinant of a square matrix: signed volume scaling under its linear map. For the affine's spatial block, $\lvert\det A_{3\times3}\rvert$ gives positive voxel volume; translation does not change it.
Softmax; e^s	Softmax converts real scores into positive weights summing to one: exponentiate each score using e^s, then divide by the sum of exponentials over the chosen axis. Slide 4 normalizes each row.
$\mathbf{1}[Q]$; $\sum_{i=1}^{n}$	The indicator is one when proposition Q holds and zero otherwise. The sum adds the indicated terms for $i = 1, \ldots, n$; division by n yields their mean.
$\lvert a\rvert$; $\lvert S\rvert$; $\lVert v\rVert^2$	Absolute value for scalar a; number of elements for finite set S; squared Euclidean length $\sum_i v_i^2$ for vector v. Context determines which bars are intended.
min; max; $[a]_+$	Minimum and maximum return values. The positive part is $[a]_+ = \max(a, 0)$, leaving positive values unchanged and replacing negative ones by zero.
$\arg\min$; $\arg\max$	Return the arguments achieving an optimum. For example, $\hat{\theta} \in \arg\min_\theta J(\theta)$ names a minimizing parameter; $\min_\theta J(\theta)$ is the minimum objective value. Ties are possible.
$\Gamma \vdash \phi$	Read "ϕ is derivable from premises Γ under the chosen formal rules." This is distinct from an empirical claim that the premises are true.
$\partial_t c$; $\dot{M}$	Partial time derivative of a space–time field, holding position fixed; ordinary time derivative dM/dt of a total. Their units include division by time.
∇c; $\nabla \cdot \mathbf{j}$	Gradient: spatial derivatives pointing toward increase. Divergence: net outward flux per local volume. For flux $\mathbf{j} = -D\nabla c$, redistribution is $-\nabla \cdot \mathbf{j}$.
$\Delta o = \nabla^2 o$	The Laplacian, summing second spatial derivatives. On slide 19 it describes oxygen smoothing. Elsewhere Δ labels a finite contrast or effect; it is then a different use of the symbol.
$\int_\Omega c\, dV$	A volume integral over domain Ω. Think of adding $c_i v_i$ over increasingly fine voxels. dV carries volume units; density times volume gives cell number.

Probability, uncertainty and decision notation

Term or symbol	Meaning and how to read it
$P(A)$; $P(A \mid B)$	Probability of event A; probability of A given B, defined as $P(A \cap B)/P(B)$ when $P(B) > 0$. Conditioning reverses neither events nor assumptions automatically.
$\neg D$; $+$; $-$	Absence of disease, a positive test and a negative test. Here D names an event; it is not the diffusion coefficient of the growth equations.
$\pi = P(D)$	Prevalence: the fraction with disease in the specified population, before conditioning on the test result. Elsewhere π can name an action policy.
Se; Sp	Sensitivity $P(+ \mid D)$ and specificity $P(- \mid \neg D)$. False-positive probability among unaffected people is $1 - \text{Sp}$.
$P(D \mid +)$	PPV: reverse the conditioning to ask about disease among positive results. It depends on prevalence as well as sensitivity and specificity.
p; C_{FP}; C_{FN}	For the decision example, p is the event probability given available evidence. The two costs penalize acting without the event and not acting despite it; correct decisions have zero cost in that example.
$\mathbb{E}[Z]$; $Z \sim P$	Expected value is the probability-weighted average of Z. The symbol $\sim$ states that a random quantity follows a named distribution.
$\text{Var}(Z)$	Variance $\mathbb{E}[(Z - \mathbb{E}Z)^2]$, in squared units of Z. The standard deviation is its square root.
$s_d/\sqrt{n}$	Estimated SE of a mean of n independent, identically distributed differences, using sample standard deviation s_d and assuming finite variance. Dependence requires a different analysis.
$Y(1), Y(0)$; Δ	Potential outcomes under two interventions and their declared contrast. Only the outcome under the assigned condition is observed for an individual.

Read Bayes' rule as an accounting identity. For disease event D,

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+)}, \qquad P(+) = \text{Se}\,\pi + (1 - \text{Sp})(1 - \pi).$$

The numerator counts true-positive probability; the denominator counts all positive probability. In the hypothetical population of 10,000, the 90 true positives and 990 false positives give $90/(90 + 990) = 1/12$. Population size cancels, leaving a dimensionless fraction.

Read the action boundary separately. Comparing $(1-p)C_{FP}$ with pC_{FN} gives threshold $C_{FP}/(C_{FP}+C_{FN})$ for positive costs. This threshold follows from the stipulated losses; neither sensitivity nor PPV chooses acceptable consequences. Slides 10 and 25 keep that distinction explicit.

Growth, oxygen, measurement and display notation

Term or symbol	Meaning and how to read it
$\mathbf{x}$; t; Ω	Physical position in mm, time in days and the tissue domain. Bold $\mathbf{x}$ denotes a position vector; elsewhere plain x can denote an image or input feature vector.
$c(\mathbf{x}, t)$; $z(\mathbf{x}, t)$	Viable density and immobile necrotic material. In the coupled model both use cell equivalents/mm^3 so their transfer and total can be accounted for consistently.
K; $u = c/K$	Constant capacity in the same density units; u is dimensionless normalized density, not a probability of cancer. Elsewhere K can denote attention keys or an iteration budget.
D; ρ	Diffusivity in mm^2/day and proliferation rate in day^{-1}. The lessons assign tissue-dependent scalar D; heterogeneity is distinct from directional anisotropy.
o; p	Dimensionless relative oxygen and delivery factor. Here $p = [1 + \alpha(c + z)/K]^{-1}$ is a perfusion rule, not the event probability p of a decision example.
ℓ_o; α; β	Oxygen smoothing length in mm; dimensionless perfusion-suppression and uptake parameters. They are assigned model settings.
θ_p; θ_d; g, h	Dimensionless oxygen thresholds and response functions, with $0 < \theta_d < \theta_p < 1$. g reduces proliferation; h controls death transfer (slide 20).
B; T; δ; λ_z	Local births and viable-to-necrotic transfer in cell equivalents/mm^3/day; death-transfer and clearance rates in day^{-1}. On slide 16, B instead labels a density integral.
$\mathbf{n} \cdot \mathbf{j} = 0$	No-flux boundary: outward unit normal $\mathbf{n}$ has zero dot product with transport flux $\mathbf{j}$. No material crosses that boundary, although births inside can change the total.
$M = \int_\Omega (c + z)\, dV$	Total viable plus necrotic material in cell equivalents. Internal transfer cancels from $\dot{M}$; births add material and clearance removes it.
A; v; N	For geometry, A is the image affine, v voxel volume in mm^3, and N an integrated cell count. Attention matrix A and sample count N are separate, locally defined uses.
B; V_τ; τ	On slide 16, density integral and threshold volume, both in mL, and dimensionless display cutoff. $B = (v/1000) \sum_i u_i$; $V_\tau = (v/1000) \sum_i \mathbf{1}[u_i \geq \tau]$.
H; S_θ; $U_\theta(t)$	An observation or display operator, as specified; a numerical solution operator; and the computed spatial fields. Subscripts on H state which display settings are fixed.
$\theta = (s, \rho)$; $J(\theta)$	For parameter recovery, diffusivity multiplier and growth rate, and mean squared observation residual. This use of s differs from a general dynamical state.

Check dimensions before interpreting a total. Multiplying density (cells/mm^3) by voxel volume (mm^3) gives cells. Summing the fraction u times voxel volume instead gives mm^3; dividing by 1000 gives mL. A rate balance for M must have cell equivalents/day on both sides. The same numerical array can support several summaries, whose names and units must remain explicit.

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